Muhammad Hani Zaheer

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EMPLOYMENT

- **Postdoctoral Researcher** **Jan 2025 - Present**
  University of Delaware | Department of Electrical & Computer Engineering
- **Graduate Research Assistant** **June 2020 – Jan 2025**
  University of Delaware | Department of Physics & Astronomy
- **Graduate Teaching Assistant** **Sep 2019 – May 2020**
  University of Delaware | Department of Physics & Astronomy

EDUCATION

- **Ph.D. Physics | University of Delaware** **2019 – 2024**
  GPA: 4.0 | Thesis: Detection Prospects of Ultralight Scalars with Quantum Sensing Experiments
- **M.Sc. Physics | University of Karachi** **2018**
  First Position | Percentage: 90.90%
- **B.Sc. (Hons.) Physics | University of Karachi** **2015 – 2017**
  First Position | Percentage: 87.86% | Double Minor: *Mathematics*, *Statistics*

TEACHING EXPERIENCE

- **Co-Instructor | PHYS 209 Aspects of Modern Physics** **Spring 2026**
  University of Delaware | Department of Physics & Astronomy
- **Guest Lecturer | QSEG 851 Advanced Topics in Quantum Information**
  - Spring 2025: 2 lectures
  - Spring 2026: 1 lecture
- **Graduate Teaching Assistant** **Sep 2019 – May 2020**
  University of Delaware | Department of Physics & Astronomy
  - PHYS 201 Introductory Physics I — Fall 2019
  - PHYS 207 Fundamentals of Physics I — Winter 2020
  - PHYS 207 Fundamentals of Physics I — Spring 2020
- **Teaching Intern | The Citizens Foundation** **Summer 2017**
  - Taught physics to underprivileged students preparing for university admissions tests as part of The Citizens Foundation Alumni Development Program.

HONORS AND AWARDS

- **Daicar-Bata Award** (2022) for Highest GPA (4.0) in Ph.D. Physics at University of Delaware.
- **DAMOP Student Participation Award** (2022) for presenting a paper at APS DAMOP.
- **Gold Medal** (2018) for Highest Percentage (90.90%) in M.Sc. Physics at University of Karachi.
- **Gold Medal** (2017) for Highest Percentage (87.86%) in B.Sc. Physics at University of Karachi.

TALKS

- **Searching for axion dark matter with magnetic resonance force microscopy** on June 2, 2026, in *APS DAMOP Meeting 2026*.
- **Quantum Sensing for Ultralight Dark Matter** on April 20, 2026, at the University of Delaware [Invited Seminar].

- **Optimizing magnetic resonance force microscopes for dark matter detection** on October 31, 2025, at *DRD5 Collaboration Meeting* at CERN [Invited Talk].
- **Searching for Axions with Magnetic Resonance Force Microscopy** on October 8, 2025, at *CPAD 2025* at Penn.
- **Quantum Metrology Algorithms for Scalar Dark Matter with Clocks** on September 1, 2025, at Quaid-i-Azam University [Invited Seminar].
- **Detection of Bosenova with Quantum Sensing Experiments** on April 3, 2024, in *APS April Meeting 2024*.
- **Quantum metrology algorithms for dark matter searches with $^{229}Th^{3+}$** on May 31, 2022, in *APS DAMOP Meeting 2022*.

PUBLICATIONS

Quantum Sensing

- E. Kashi, **M.H. Zaheer**, R. Petery, and S. Singh, Searching for axion dark matter with magnetic resonance force microscopy, *arXiv*:2512.12120 (2025). Accepted for publication in *Phys. Rev. D*.
- **M.H. Zaheer**, N.J. Matjelo, D.B. Hume, M.S. Safronova, and D.R. Leibrandt, Quantum metrology algorithms for dark matter searches with clocks, *Phys. Rev. A*, 111, 012601 (2025).

Dark Matter Phenomenology

- J. Arakawa, **M.H. Zaheer**, V. Takhistov, M.S. Safronova, J. Eby, and C. Cheung, Multimessenger Astronomy Beyond the Standard Model: New Window from Quantum Sensors, *J. Cosmol. Astropart. P.*, 2026 (02), 026 (2026).
- J. Arakawa, J. Eby, M. Safronova, V. Takhistov, and **M.H. Zaheer**, Detection of Bosenovae with Quantum Sensors on Earth and in Space, *Phys. Rev. D,* 110, 075007 (2024).
- J. Arakawa, **M.H. Zaheer**, J. Eby, V. Takhistov, and M. Safronova, Bosenovae with Quadratically-Coupled Scalars in Quantum Sensing Experiments, *J. High Energy Phys.*, 2024, 222 (2024).
- D. Antypas, et al., New horizons: scalar and vector ultralight dark matter, *arXiv*:2203.14915 (2022). Submitted to Snowmass 2021.

Physics Education

- A. Razzak, **M.H. Zaheer**, M.B. Khan, and Z. Uddin, Root Finding Methods Through GUI in Spreadsheets: Root Finding Methods Through GUI in Spreadsheets, *Proc. Pak. Acad. Sci. A*, 59(1), 37 (2022).
- Z. Uddin, and **M.H. Zaheer**, Simulating Physics Experiments in Spreadsheets–Experimenting with Ohm's Law, *Phys. Teach.*, 57(3), 182 (2019).

Other Publications

- **M.H. Zaheer**, M.B. Khan, S.M. Iqbal, and Z. Uddin, A Python Program for Computation of Transition Probabilities, Oscillator Strengths for Li-like ions, *arXiv*:2405.19663 (2024).
- R. Siddiq, M.N. Hameed, **M.H. Zaheer**, M.B. Khan, and Z. Uddin, Rydberg energies and transition probabilities of Li I for np–ms ($m \leq 5$) transitions, *BJBAS*, 11(1), 42 (2022).
- Z. Uddin, M.B. Khan, **M.H. Zaheer**, W. Ahmad, and M.A. Qureshi, An alternate method of evaluating Lagrange multipliers of MEP, *SN App. Sci.*, 1, 1 (2019).

MEDIA COVERAGE

- Article on "Detection of Bosenovae with Quantum Sensors on Earth and in Space" in Live Science on July 25, 2023 by Dr. Paul Sutter titled "Invisible supernovas called 'bosenovas' may be exploding all around us, new research suggests".

SERVICE AND OUTREACH

- Referee for *Physica Scripta*, *The Physics Teacher* and *European Journal of Physics*.

- Organized and led several iterations of Quantum Optics Journal Club at University of Delaware.
- Founding member of Pakistan Graduate Student Association (PGSA) at the University of Delaware and served as Outreach Officer from 2021-2022.
- Taught Physics for university admission test preparation to underprivileged students in a Teaching Internship for The Citizens Foundation in 2017.
- Helped organize *National Conference on Material Science* (March 27-28, 2018) at the University of Karachi.
- Helped organize *National Conference on Multi-Disciplinary Topics in Physics* (Feb 12-13, 2016), organized by Karachi Physical Society at the University of Karachi.